



RV Institute of Technology and Management ®

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**Department of Electronics and Communication**

**Engineering**



**Course Name: INTRODUCTION TO CYBER SECURITY**

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## **INTRODUCTION: CYBER FORENSICS**

### **CYBER FORENSICS:**

Computer forensics is the application of investigation and analysis techniques to gather and preserve evidence.

Forensic examiners typically analyze data from personal computers, laptops, personal digital assistants, cell phones, servers, tapes, and any other type of media. This process can involve anything from breaking encryption, to executing search warrants with a law enforcement team, to recovering and analyzing files from hard drives that will be critical evidence in the most serious civil and criminal cases.

The forensic examination of computers, and data storage media, is a complicated and highly specialized process. The results of forensic examinations are compiled and included in reports. In many cases, examiners testify to their findings, where their skills and abilities are put to ultimate scrutiny.

### **Historical Background of Cyber Forensics**

Cyber forensics, also known as digital forensics, has its roots in the early days of computer technology. As personal computers became more common in the 1980s, law enforcement agencies began to encounter cases that involved digital evidence.

The first use of digital evidence in a criminal trial was in 1984, when a hacker was prosecuted for breaking into a computer system. The case highlighted the need for law enforcement to develop specialized techniques and tools for investigating and analyzing digital evidence.

In the following years, cybercrime became more prevalent, and law enforcement agencies began to establish specialized units to investigate digital crimes. These early cybercrime units focused on the investigation of computer intrusions, hacking, and viruses.

The 1990s saw the emergence of the World Wide Web and the widespread use of the internet, leading to a surge in cybercrime. This prompted the development of more advanced tools and techniques for digital forensics, including specialized software and hardware.

The turn of the century saw a further increase in cybercrime, including the rise of new types of crimes such as identity theft and online fraud. In response, governments around the world began to



enact laws and regulations to combat cybercrime, and the field of cyber forensics continued to evolve to meet these new challenges.

Today, cyber forensics has become an essential part of law enforcement and corporate security, with experts in the field playing a critical role in the investigation and prosecution of cybercrime. The field continues to evolve rapidly, driven by advances in technology and the ongoing development of new types of cyber threats.

### **DIGITAL FORENSICS SCIENCE:**

Digital Forensics is defined as the process of preservation, identification, extraction, and documentation of computer evidence which can be used by the court of law. It is a science of finding evidence from digital media like a computer, mobile phone, server, or network. It provides the forensic team with the best techniques and tools to solve complicated digital-related cases.

Digital Forensics helps the forensic team to analyze, inspect, identify, and preserve the digital evidence residing on various types of electronic devices.

Digital forensic science is a branch of forensic science that focuses on the recovery and investigation of material found in digital devices related to cybercrime.

The field of digital forensic science plays a critical role in the investigation and prosecution of cybercrime, as well as in other areas of cybersecurity. Some of the key roles of digital forensic science include:

1. **Preservation of Digital Evidence:** Digital forensic experts are responsible for preserving digital evidence in a legally admissible manner. This involves ensuring that data is not tampered with, and that a chain of custody is established to document the handling of evidence.
2. **Collection of Digital Evidence:** Digital forensic experts use specialized tools and techniques to collect data from computers, smartphones, and other digital devices. This data can include emails, instant messages, and other forms of communication, as well as files and other data stored on the device.
3. **Analysis of Digital Evidence:** Digital forensic experts use a range of techniques to analyze digital evidence, including data recovery, data carving, and forensic imaging. The aim of this analysis is to extract as much information as possible from the data, and to use it to identify the perpetrator of a cyber attack or other digital crime.



4. **Presentation of Digital Evidence:** Digital forensic experts must present their findings in a clear and concise manner, often in a court of law. This may involve preparing detailed reports, creating forensic charts and diagrams, and providing expert testimony.
5. **Development of Best Practices:** Digital forensic experts are also involved in the development of best practices for cybersecurity. This may include the development of guidelines for the collection and analysis of digital evidence, as well as the development of strategies for the protection of sensitive data and the prevention of cyber attacks.

In summary, the roles of digital forensic science include the preservation, collection, analysis, and presentation of digital evidence, as well as the development of best practices for cybersecurity. These roles are critical in the investigation and prosecution of cybercrime, and in the protection of sensitive data and computer systems from attack.

Digital forensic science involves the application of forensic techniques to digital devices and data in order to investigate and gather evidence related to criminal activities or other security incidents. Here are some of the different scenarios where digital forensic science may be used:

1. **Cybercrime Investigations:** Digital forensic techniques are used to investigate cybercrime such as hacking, identity theft, online fraud, and other computer-related crimes.
2. **Intellectual Property Theft:** Digital forensic techniques can be used to investigate cases of intellectual property theft, such as the unauthorized copying and distribution of copyrighted materials or trade secrets.
3. **Employee Misconduct:** Digital forensic techniques can be used to investigate cases of employee misconduct, such as theft of company data, violation of company policies, or breach of contract.
4. **Data Recovery:** Digital forensic techniques can be used to recover lost, damaged or deleted data from digital devices.
5. **Incident Response:** Digital forensic techniques can be used to investigate security incidents and determine the extent of the damage, identify the source of the attack, and develop strategies to mitigate the damage.
6. **Litigation Support:** Digital forensic techniques can be used to provide evidence in a court of law, such as in civil litigation, criminal trials, or administrative hearings.
7. **Compliance Audits:** Digital forensic techniques can be used to conduct compliance audits, such as ensuring that data privacy and security standards are being met, and to identify vulnerabilities that need to be addressed.

In summary, digital forensic science can be used in a wide range of scenarios to investigate and gather evidence related to cybercrime, intellectual property theft, employee misconduct, data recovery, incident response, litigation support, and compliance audits.



## **THE NEED FOR COMPUTER FORENSICS**

Computer forensics, also known as digital forensics, is a critical component of cybersecurity that is necessary for the investigation and prosecution of cybercrime. Here are some of the key reasons for the need for computer forensics:

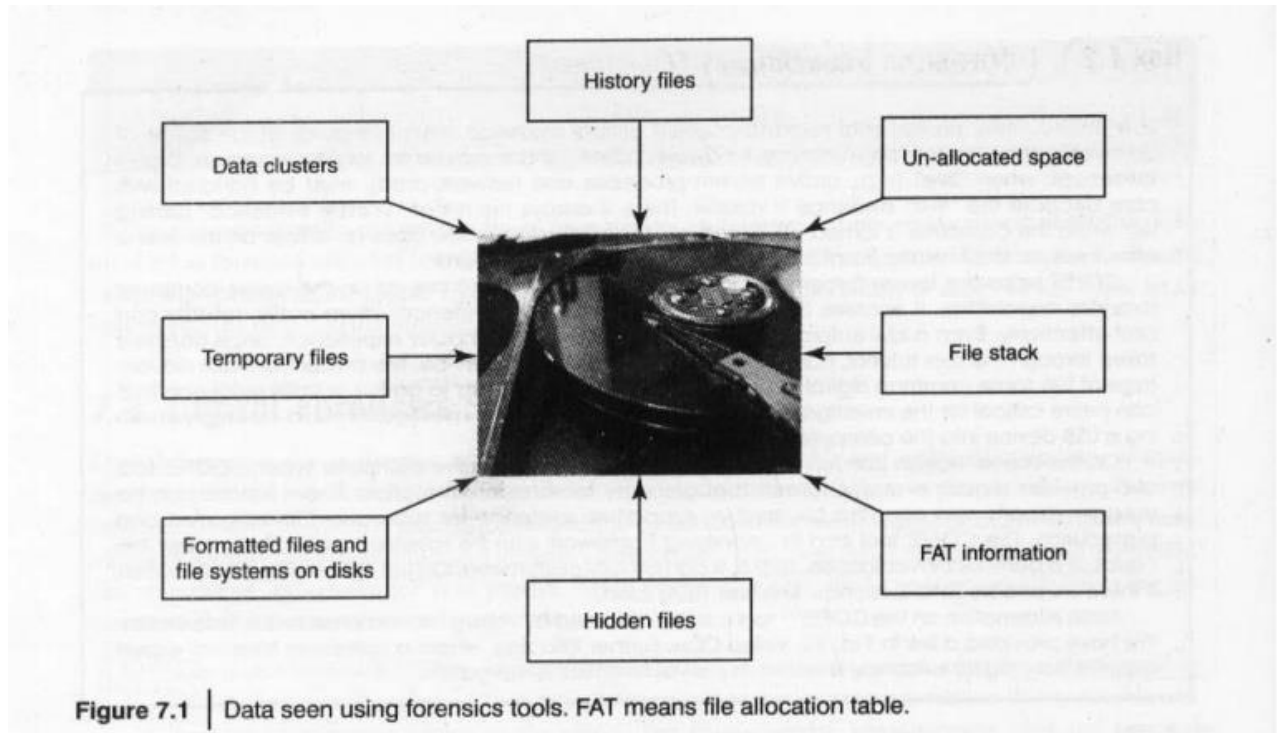
1. **Identification of Cybercriminals:** Cybercriminals often use sophisticated techniques to hide their identities and cover their tracks. Digital forensic tools and techniques can help investigators to identify the perpetrators of cybercrime and gather evidence for prosecution.
2. **Collection of Digital Evidence:** Digital evidence, including data stored on hard drives, smartphones, and other digital devices, is often essential in cybercrime investigations. Digital forensic techniques can be used to collect and preserve this evidence in a legally admissible manner.
3. **Analysis of Digital Evidence:** Digital forensic experts use specialized tools and techniques to analyze digital evidence and uncover details about the methods used by cybercriminals. This can help investigators to determine the extent of the attack and identify vulnerabilities that need to be addressed to prevent future attacks.
4. **Recovery of Lost Data:** Digital forensic techniques can be used to recover lost or deleted data, even if it has been overwritten or damaged. This can be particularly valuable in cases where data has been deliberately deleted or destroyed as part of a cyber attack.
5. **Protection of Sensitive Data:** Digital forensic techniques can also be used to identify vulnerabilities in computer systems and networks, and to develop strategies to protect sensitive data from cyber attacks.

In summary, computer forensics is essential for the investigation and prosecution of cybercrime, the collection and analysis of digital evidence, the recovery of lost data, and the protection of sensitive information. As cybercrime continues to grow in scope and complexity, the importance of digital forensic expertise is only expected to increase.

## **CYBER FORENSICS AND DIGITAL EVIDENCE:**

Digital evidence is information stored or transmitted in binary form that may be relied on in court. It can be found on a computer hard drive, a mobile phone, among other places. Digital evidence is commonly associated with electronic crime, or e-crime, such as child pornography or credit card fraud. However, digital evidence is now used to prosecute all types of crimes, not just e-crime. For example, suspects' e-mail or mobile phone files might contain critical evidence regarding their intent, their whereabouts at the time of a crime and their relationship with other suspects. In 2005, for example, a floppy disk led investigators to the BTK serial killer who had eluded police capture since 1974 and claimed the lives of at least 10 victims.

In an effort to fight e-crime and to collect relevant digital evidence for all crimes, law enforcement agencies are incorporating the collection and analysis of digital evidence, also known as computer forensics, into their infrastructure. Law enforcement agencies are challenged by the need to train officers to collect digital evidence and keep up with rapidly evolving technologies such as computer operating systems.



Using digital forensic techniques, one can analyze and recover digital evidence from various types of digital devices and media, such as computers, smartphones, hard drives, and cloud storage. This evidence can be used to investigate and solve various types of crimes, including cybercrime, fraud, and terrorism. Digital forensic techniques can also be used to verify the authenticity of digital data and to ensure compliance with legal and regulatory requirements.

The evidence collection phase is a critical step in the cyber forensics process. The following are some general guidelines for evidence collection in cyber forensics:

1. **Act quickly:** Evidence can be easily destroyed or overwritten, so it's important to collect digital evidence as soon as possible to prevent loss or alteration of data.
2. **Document everything:** Document all actions and observations made during the collection process. This includes noting the time and date of collection, the name of the person who collected the evidence, and any actions taken during the collection process.
3. **Minimize changes:** Minimize any changes to the digital device during the collection process. This includes not turning off the device, not running any applications, and not changing any settings.

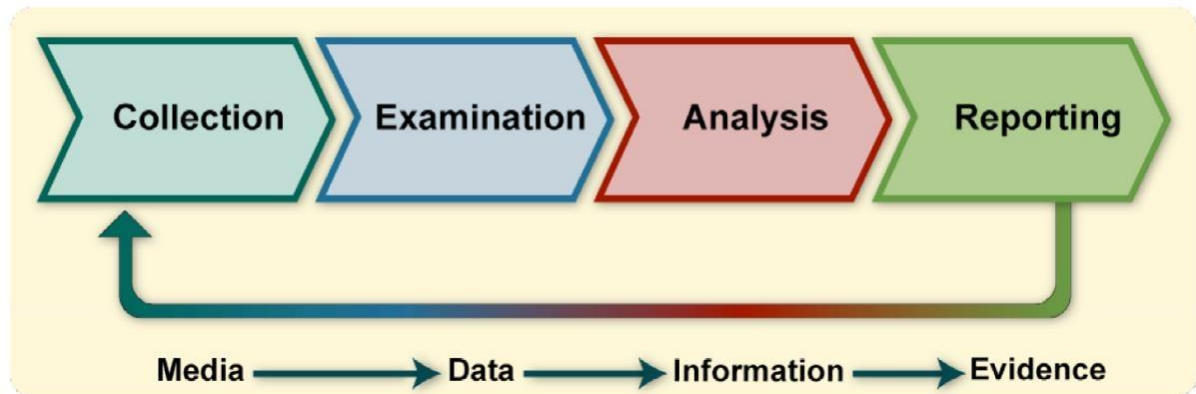


4. **Protect the evidence:** Use write-blockers or other tools to prevent the device from being written to or altered during the collection process. Store the digital evidence in a secure location to prevent any tampering or unauthorized access.
5. **Create an image:** Create an exact copy of the device or media being analyzed using a forensic tool, such as a disk imaging tool, to preserve the original data.
6. **Verify the image:** Verify the integrity of the image by creating a hash value or digital signature. This can be used to ensure that the image has not been altered during the collection process.
7. **Use multiple methods:** Use multiple methods to collect digital evidence, including collecting data from different devices or sources, and using different forensic tools or methods to ensure the completeness and accuracy of the evidence.
8. **Follow legal and ethical guidelines:** Adhere to all legal and ethical guidelines for evidence collection, including obtaining proper authorization and consent, and respecting the privacy rights of individuals involved.

By following these guidelines, cyber forensic investigators can ensure the integrity and admissibility of digital evidence in court, while also protecting the rights and privacy of individuals involved.



## DIGITAL FORENSICS LIFECYCLE:



**Collection:** The first step in the forensic process is to identify potential sources of data and acquire data from them.

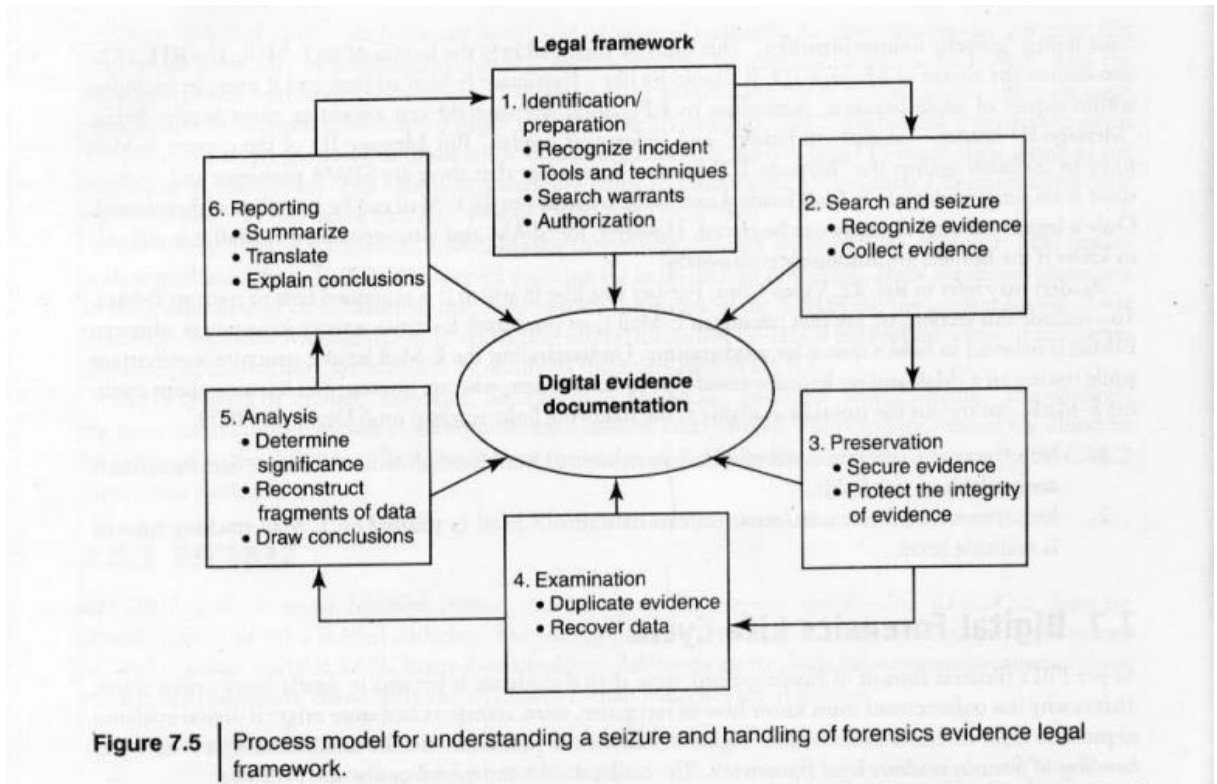
**Examination:** After data has been collected, the next phase is to examine the data, which involves assessing and extracting the relevant pieces of information from the collected data. This phase may also involve bypassing or mitigating OS or application features that obscure data and code, such as data compression, encryption, and access control mechanisms.

**Analysis:** Once the relevant information has been extracted, the analyst should study and analyze the data to draw conclusions from it. The foundation of forensics is using a methodical approach to reach appropriate conclusions based on the available data or determine that no conclusion can yet be drawn.

**Reporting:** The process of preparing and presenting the information resulting from the analysis phase. Many factors affect reporting, including the following:

- a. **Alternative Explanations:** When the information regarding an event is incomplete, it may not be possible to arrive at a definitive explanation of what happened. When an event has two or more plausible explanations, each should be given due consideration in the reporting process. Analysts should use a methodical approach to attempt to prove or disprove each possible explanation that is proposed.
- b. **Audience Consideration.** Knowing the audience to which the data or information will be shown is important.
- c. **Actionable Information.** Reporting also includes identifying actionable information gained from data that may allow an analyst to collect new sources of information





## Different Phases involved in Computer Forensics

### Preparing for the evidence and identifying the evidence

To prepare for the evidence, you should take the following steps:

1. Establish a forensics lab or secure the crime scene.
2. Determine the scope of the investigation and identify all the potential sources of evidence.
3. Develop an evidence collection plan.
4. Assemble the necessary equipment and tools for the investigation.
5. Document the chain of custody of the evidence.

To identify the evidence, you should follow these steps:

1. Conduct a preliminary investigation to identify potential sources of evidence.
2. Determine the type of evidence that might be relevant to the investigation.
3. Search for evidence using various digital forensic tools and techniques.
4. Collect and preserve the evidence following standard procedures.
5. Continuously validate and verify the authenticity and integrity of the evidence to ensure its admissibility in court.



### **Collection and Recording**

The collection and recording phase is a critical step in computer forensics. During this phase, digital evidence is collected and recorded in a forensically sound manner to ensure that it is admissible in a court of law. The following steps are typically involved:

1. Identify the sources of digital evidence: This can include computer systems, mobile devices, network devices, cloud storage, or any other storage devices that may contain relevant data.
2. Secure the crime scene: The area where the digital evidence is located must be secured to prevent any further tampering or alteration of the evidence.
3. Follow proper evidence handling procedures: Collecting digital evidence must be performed in a forensically sound manner to ensure the integrity and authenticity of the evidence.
4. Record all relevant information: It's crucial to document everything related to the evidence, including the location, time, date, and people involved in the collection process.
5. Make a bit-for-bit copy of the original media: It's essential to make a forensically sound copy of the original media to preserve the integrity of the evidence. This copy is used for analysis and must be identical to the original media.
6. Hash the evidence: Hashing is a process that generates a unique fingerprint of the evidence. This fingerprint is used to validate the authenticity and integrity of the evidence and is recorded in a chain of custody log.
7. Store the evidence securely: Digital evidence must be stored in a secure and controlled environment to prevent unauthorized access or tampering.

### **Storing and Transporting**

The storing and transporting phase in computer forensics involves securely storing and transporting the digital evidence to ensure that it remains intact and admissible. This phase is critical to maintaining the integrity of the evidence and ensuring that it can be presented in a court of law. The following steps are typically involved:

1. Choose an appropriate storage location: The storage location must be secure, controlled, and have limited access. The evidence must be protected from physical damage, theft, and environmental factors.
2. Label and document the evidence: Each piece of evidence must be labeled and documented, including the case number, item number, and description. This information is used to maintain the chain of custody.
3. Implement access controls: Only authorized personnel should be granted access to the evidence.

Access must be documented including who accessed the evidence when and for what reason



4. Store the evidence securely: Digital evidence must be stored in a secure and controlled environment to prevent unauthorized access or tampering.
5. Transport the evidence securely: When transporting the evidence, it must be protected from physical damage, theft, and environmental factors. The chain of custody must be maintained during transportation, and access must be restricted to authorized personnel.
6. Document the transport process: The transport process must be documented, including the date, time, and location of the transfer, the name of the transporter, and the method of transportation.
7. Confirm the integrity of the evidence: When the evidence reaches its destination, it must be confirmed that it is still intact and has not been tampered with. This confirmation must be documented in the chain of custody log.

### **Examination/Investigation**

The investigation phase in computer forensics is where the collected digital evidence is analyzed to reconstruct events, determine how the incident occurred, and identify the individual(s) responsible. The following steps are typically involved:

1. Analysis of digital evidence: All digital evidence collected during the collection phase is analyzed using various forensic tools and techniques, such as disk and memory analysis, network and database analysis, and mobile device analysis.
2. Reconstruction of events: By examining the digital evidence, investigators can piece together a timeline of events leading up to the incident, including who was involved and what actions were taken.
3. Correlation of evidence: Investigators analyze the various pieces of evidence collected to identify links between different events or individuals.
4. Interpretation of results: Based on the findings of the investigation, investigators can formulate conclusions and interpret the meaning of the digital evidence.
5. Documentation: All findings, conclusions, and interpretations are documented in a report that details the investigation process, the results, and the conclusions reached.
6. Presentation of findings: The findings are presented in a clear and concise manner, typically to a client or in a court of law. The presentation should be well-documented, verifiable, and admissible as evidence.
7. Reiteration: The investigators may need to repeat some or all of the investigation process, depending on the results of the analysis or the feedback from the client or the court.



### **Analysis, Interpretation and Attribution**

The Analysis, Interpretation, and Attribution phase is a crucial stage in computer forensics, where digital evidence is analyzed, interpreted, and used to attribute the crime or incident to a specific individual or group. The following are the steps involved:

1. **Analysis of digital evidence:** All digital evidence is analyzed to identify patterns, connections, and relationships between different pieces of data.
2. **Interpretation of results:** Based on the analysis, investigators can form conclusions and interpret the meaning of the digital evidence.
3. **Attribution:** Investigators use the conclusions and interpretations to attribute the crime or incident to a specific individual or group.
4. **Verification:** The conclusions and attributions made must be verified to ensure their accuracy and validity.
5. **Report writing:** Investigators must document their findings, conclusions, and attributions in a report that is clear, concise, and understandable to stakeholders.
6. **Presentation of findings:** The report is presented to stakeholders, which can include the client, law enforcement agencies, and courts.
7. **Testimony:** Forensic investigators may be required to provide testimony in court to defend their findings and conclusions. They may be required to present evidence and explain their forensic methods and techniques to the judge and jury.
8. **Reiteration:** In some cases, further investigation may be required based on feedback from stakeholders or new evidence that emerges.

Overall, the Analysis, Interpretation, and Attribution phase is critical to the success of any computer forensic investigation, as it provides valuable insights that can be used to identify and prosecute perpetrators of digital crimes.

### **Reporting**

The reporting phase in computer forensics is where the findings of the investigation are documented and presented to stakeholders. The following are the steps involved:

1. **Create a comprehensive report:** A detailed report is created that includes the scope and objectives of the investigation, the methods and techniques used, the digital evidence collected, and the findings, conclusions, and recommendations.
2. **Organize the report:** The report is organized in a clear and logical manner that makes it easy to understand and follow.
3. **Use clear and concise language:** The report is written in clear and concise language that is easily understood by both technical and non-technical stakeholders.



4. Include technical details: The report should include technical details that support the findings and conclusions of the investigation, including information about the forensic tools and techniques used.
5. Provide recommendations: The report should provide recommendations for improving security or preventing similar incidents from occurring in the future.
6. Ensure the report is accurate and verifiable: The report should be accurate and verifiable, and the sources of all data and information should be properly cited.
7. Present the report: The report is presented to the relevant stakeholders, which may include the client, law enforcement agencies, and courts.
8. Answer questions: The investigator should be prepared to answer questions from stakeholders and provide additional information to support the findings and conclusions in the report.

The reporting phase is critical to the success of the investigation, as it provides a clear and concise record of the findings and recommendations. A well-written report can also be used as evidence in court, and can help stakeholders to make informed decisions about the incident and how to prevent similar incidents from occurring in the future.

### **Testifying**

The Testifying Phase in Computer Forensics is where forensic investigators are called upon to present their findings, methods, and analysis in a court of law. The following are the steps involved in this phase:

1. Preparation: Forensic investigators must prepare for their testimony by reviewing their report, refreshing their memory, and familiarizing themselves with any legal procedures and rules of evidence.
2. Review: The investigators should review the evidence and their findings to ensure that they are fully prepared to answer questions and provide expert testimony.
3. Anticipate potential challenges: Forensic investigators should anticipate potential challenges or objections to their findings and analysis and be prepared to address them.
4. Testify in court: Forensic investigators must present their findings and conclusions in a clear and concise manner, and answer any questions posed by the judge, lawyers, and other stakeholders.
5. Provide technical explanations: The forensic investigator may be required to provide technical explanations to the court, lawyers, or the jury, in order to help them understand complex technical concepts.
6. Be honest and impartial: Forensic investigators must be honest and impartial in their testimony, and avoid speculation or making assumptions outside of their area of expertise.
7. Remain calm and professional: Testifying in court can be stressful, and forensic investigators must remain calm and professional throughout the process.



8. Be available for follow-up questions: After their testimony, forensic investigators may be required to answer follow-up questions from the court, lawyers, or other stakeholders.

The Testifying Phase is an important part of computer forensics, as it helps to present the findings and analysis of the investigation in a legal context. The testimony of forensic investigators can be used to help the court make informed decisions and ensure that justice is served.

### **Precautions to be taken when collecting electronic evidences**

When collecting electronic evidence in computer forensics, it is essential to take certain precautions to ensure that the evidence is collected in a forensically sound manner and can be used in court. The following are some of the key precautions to be taken during the collection of electronic evidence:

1. Avoid Contamination: Avoiding contamination of the evidence is crucial. The investigator should make sure that the collection process does not alter or damage the original evidence. The collection should be done in a sterile environment and with sterile tools.
2. Documentation: Proper documentation is crucial. Each step of the collection process should be properly documented. This includes who collected the evidence, where and when it was collected, and any other relevant information.
3. Chain of Custody: Maintaining the chain of custody is crucial. It ensures that the evidence remains intact, and no tampering has occurred. Each person who handles the evidence should document their involvement and sign off on the chain of custody.
4. Preservation: Preservation of the evidence is important. This includes making a copy of the data, maintaining the original evidence, and making sure that the data is stored in a secure location.
5. Proper Handling: Proper handling of the evidence is crucial. The investigator should make sure that the evidence is handled with care and that it is not damaged during the collection process.
6. Legal Requirements: The collection of electronic evidence must comply with legal requirements. The investigator must have the necessary permissions and follow the relevant legal procedures.
7. Expertise: The collection of electronic evidence requires expertise. The investigator should have the necessary technical knowledge and experience to ensure that the evidence is collected in a forensically sound manner.

Overall, collecting electronic evidence in computer forensics requires taking several precautions to ensure that the evidence is collected in a forensically sound manner and can be used in court. The above-mentioned precautions help to ensure that the evidence is collected in a way that is admissible in court and that the integrity of the evidence is maintained.



## **CHAIN OF CUSTODY CONCEPT:**

Chain of custody is a critical concept in cyber forensics, which refers to the documentation of the movement of digital evidence from the time it is collected until it is presented in court. The purpose of the chain of custody is to maintain the integrity and admissibility of the evidence by documenting who had access to the evidence, when and why it was accessed, and how it was stored and protected.

The chain of custody begins at the point of evidence collection, where the digital evidence is first identified, collected, and documented. At this stage, the investigator must document the time and date of collection, the location, the individuals present, and any actions taken during the collection process.

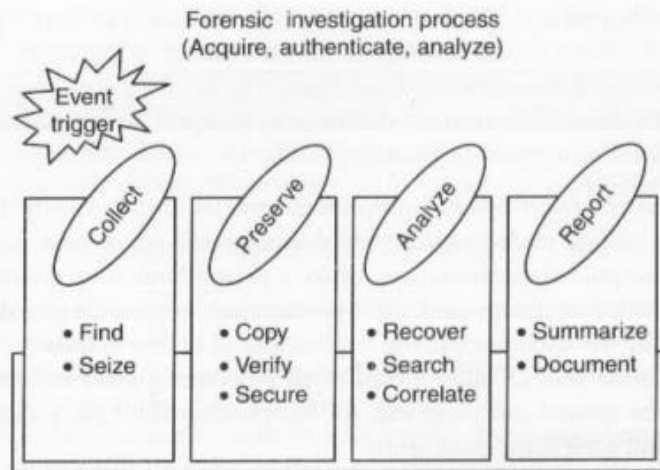
Once the evidence is collected, it must be properly stored and protected to prevent any tampering or loss of data. This involves using write-blockers or other tools to prevent the device from being written to or altered during the collection process, and storing the evidence in a secure location with limited access.

Throughout the investigation, the chain of custody must be documented and maintained to ensure the integrity of the evidence. This involves documenting each step of the process, including any transfers of the evidence, any access to the evidence, and any changes to the evidence.

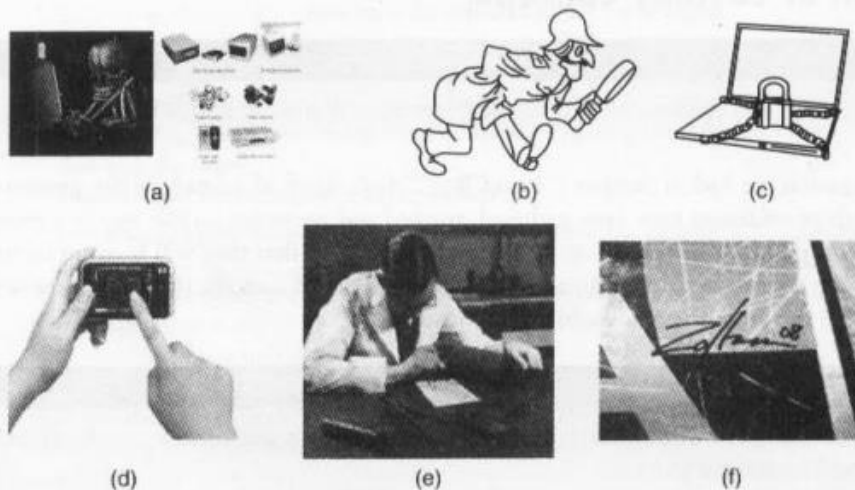
The chain of custody is particularly important when digital evidence is presented in court. The prosecution must be able to demonstrate that the evidence has not been tampered with or altered in any way, and that the evidence presented in court is the same evidence that was collected during the investigation. The defense may challenge the chain of custody if there are any gaps or inconsistencies in the documentation, which can impact the admissibility of the evidence in court.

In summary, the chain of custody is a critical concept in cyber forensics, which is used to maintain the integrity and admissibility of digital evidence by documenting the movement of the evidence from the time it is collected until it is presented in court. Proper documentation and maintenance of the chain of custody is essential to ensure that the evidence is not compromised and that it is admissible in court.





**Figure 7.10** | Maintaining chain of custody – 1.



**Figure 7.11** | Maintaining chain of custody – 2. (a) Source of evidence – where did it come from? (b) Who found it? (c) Where was it stored/locked up? (d) Who touched it/tampered with it? (e) What did they do to it? What did they do with it? (f) Human signature is always required.

## Network Forensics

Network forensics is the process of collecting, analyzing, and interpreting digital evidence from computer networks to investigate security incidents, cybercrime, or network breaches. It involves the capture and analysis of network traffic to identify malicious activities and gather evidence of the network-related activities.

The main goal of network forensics is to reconstruct the activities of an attacker, identify the attack's origin, and determine the extent of the damage. Network forensics is used by network administrators, incident response teams, and law enforcement agencies to investigate network security breaches and cyber attacks.



The process of network forensics involves the following steps:

1. **Collection of network data:** In this step, the network data is captured and stored for further analysis. Various tools, such as network sniffers, network intrusion detection systems (NIDS), and network security monitoring tools, can be used to collect the network data.
2. **Network data analysis:** In this step, the collected network data is analyzed to identify malicious activities and other network anomalies. The data is examined to determine the source and destination of the traffic, the type of traffic, and any other patterns or signatures that may be indicative of an attack.
3. **Data reconstruction:** In this step, the reconstructed data is analyzed to determine the attacker's goals, techniques, and the extent of the damage caused.
4. **Reporting:** Finally, a report is prepared summarizing the findings and recommendations for improving network security.

Network forensics can be used to investigate a wide range of network-based security incidents, including:

- Network intrusions and attacks
- Denial of Service (DoS) attacks
- Malware infections
- Phishing and other social engineering attacks
- Insider threats and data theft
- Data breaches